



MA 510: Climate Change Analysis

Course Code	MA 510
Course Name	Climate Change Analysis
L-T-P-C	3-0-0-3
Prerequisites	UG None
Intended for	UG/PG
Mutual Exclusion	None
Approval	9th Senate (May 2016)

Course Contents

- **Statistical concepts in climate research: 20 hrs**
- Misuses of statistics in climate research: testing hypotheses suggested by the data; serial correlation; using statistical recipes as "black-box" tools;
 - Hypothesis testing Type I and Type II errors, significance, power, etc; historical developments and controversy around classical statistical significance test and its interpretation.
 - Basics of Bayesian statistics: Introduction to Bayesian statistics; Bayesian climate change assessment.
 - Advanced data analysis technique like functional data analysis and wavelet analysis to detect climate changes is discussed in detail.
- **Detection and attribution of anthropogenic climate change: 10 hrs**
 - To identify the causes of recent observed climate variations
 - To evaluate the performance of climate models in simulating the observed climate variations over the last century
 - To constrain the projections of future climate change
- **Climate science beyond the IPCC 6 hrs**
 - Univariate analysis of global mean temperature comparing change with internal variability
 - Study of different causes that affect global radiation balance; increasing greenhouse gases, increasing solar irradiance
- **Use the spatial pattern of the temperature response to differentiate between different causes and fingerprint analysis Minor project/seminars 6 hrs**

Text Books

1. Montgomery, D., Jennings, C.L. and Kulahci, M., **Introduction to Time Series Analysis and Forecasting**, Wiley-Interscience, 2008.
2. Chatfield, C., **The Analysis of Time Series**, 6th Edition Chapman & Hall/CRC, 2004.

References

1. Robert H. Shumway and David S. Stoffer, **Time Series Analysis and Its Applications with R Examples**, 3rd edition, Springer Texts in Statistics, 2006.
2. Hans von Storch, Francis W. Zwiers, **Statistical Analysis in Climate Research**, Cambridge University Press (2002)
3. Hartmann U., Ramirez F., **Real Time Detection of Turning Points in Financial Time Series**, GRIN Verlag, 2013.